

SECTION 1.4

GRAPHICAL METHODS
FOR DESCRIBING
QUANTITATIVE DATA

The most popular and traditional graphical method for describing quantitative data is the relative frequency histogram (often called a relative frequency distribution). For example, suppose we want to describe graphically the data shown in Table 1.1. The CPU times given in the table are the amounts of time (in seconds) 25 jobs were in control of a large mainframe computer's central processing unit (CPU). These 25 values represent a sample selected from the 1,000 CPU times listed in Appendix IV. The steps to follow in constructing a relative frequency histogram are listed in the box. The results for the data of Table 1.1 are shown in Table 1.2.

STEPS TO FOLLOW IN CONSTRUCTING A RELATIVE FREQUENCY HISTOGRAM

Step 1. Calculate the range of the data:

$$\text{Range} = \text{Largest observation} - \text{Smallest observation}$$

The range for the data of Table 1.1 is

$$\text{Range} = 4.75 - 0.02 = 4.73$$

Step 2. Divide the range into between five and twenty classes of equal width. The number of classes is arbitrary, but you will obtain a better graphical description if you use a small number of classes for a small amount of data and a larger number of classes for large data sets. The lowest (or first) class boundary should be located below the smallest measurement, and the class width should be chosen so that no observation can fall on a class boundary. We will use seven classes for the data of Table 1.1. Then the approximate class width is

$$\text{Approximate class width} = \frac{\text{Range}}{7} = \frac{4.73}{7} = .676$$

We will round this width upward and use a class width of .7. The first class will be located at .015, just below the smallest CPU time. The resulting seven class intervals are shown in Table 1.2.

Step 3. For each class, count the number of observations that fall in that class. This number is called the class frequency.

Step 4. Calculate each class relative frequency:

$$\text{Class relative frequency} = \frac{\text{Class frequency}}{\text{Total number of measurements}}$$

The class frequencies and relative frequencies for the data of Table 1.1 are shown in columns 4 and 5, respectively, of Table 1.2.

Step 5. The relative frequency histogram is essentially a bar graph in which the categories are the classes. The relative frequency histogram for the data of Table 1.1 is shown in Figure 1.4.

TABLE 1.1

A Sample of $n = 25$
Job CPU Times (in
seconds) Selected
from Appendix IV

1.17	1.61	1.16	1.38	3.53
1.23	3.76	1.94	0.96	4.75
0.15	2.41	0.71	0.02	1.59
0.19	0.82	0.47	2.16	2.01
0.92	0.75	2.59	3.07	1.40

TABLE 1.2

Calculating Class Relative
Frequencies for the Data
of Table 1.1

CLASS	CLASS INTERVAL	DATA TABULATION	CLASS FREQUENCY	CLASS RELATIVE FREQUENCY
1	.015-.715		5	.20
2	.715-1.415		9	.36
3	1.415-2.115		4	.16
4	2.115-2.815		3	.12
5	2.815-3.515		1	.04
6	3.515-4.215		2	.08
7	4.215-4.915		1	.04
			Totals $n = 25$	1.00

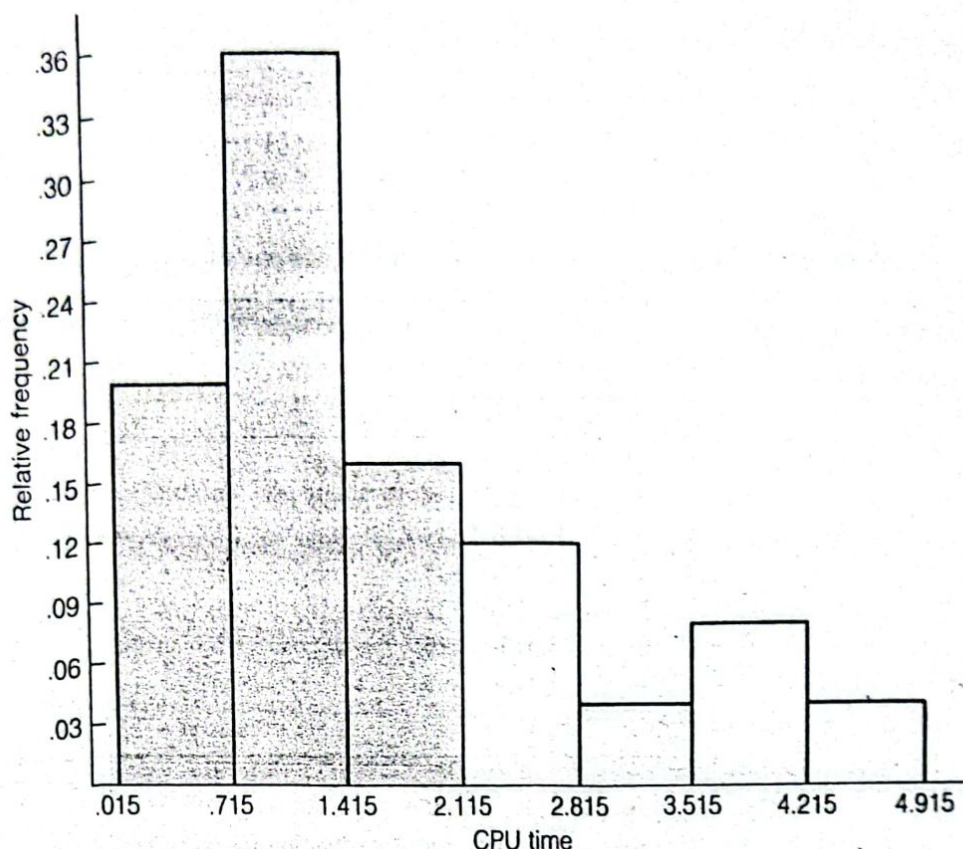
Since the bars in a relative frequency histogram are of equal width, the area of a particular bar is proportional to the corresponding class relative frequency. If we let the total area of the bars equal 1, then the area of a particular bar is equal to its corresponding class relative frequency. Furthermore, if we select one observation from among the $n = 25$ observations in Table 1.1, then the probability that the observation will fall in a particular class is equal to the relative frequency of that class. The probability that the observation will fall in one of two or more specific classes is equal to the sum of their respective relative frequencies and is proportional to the total area of the bars corresponding to those classes. For example, the probability that the observation will be a CPU time less than 2.115 seconds is equal to .72, the sum of the relative frequencies for classes 1, 2, and 3 of Table 1.2. This probability is proportional to the shaded area of Figure 1.4 on page 14.

[Note: Most graphical descriptions of large data sets, whether qualitative or quantitative, are performed on a computer by one of several easy-to-use statistical computer program packages. In Chapter 2 we will show you how to enter data into a computer for use with each of four such packages: BMDP, Minitab, SAS, and SPSS*. In Sections 2.6 and 2.7, we will show you how to use the programs to construct a relative frequency histogram.]

Another graphical method for describing quantitative data is the stem and leaf display, which is widely used in exploratory data analysis when the data set is small. The next box contains the steps to follow in constructing a stem and leaf display for the 25 CPU times of Table 1.1.

FIGURE 1.4

Relative Frequency
Histogram for the $n = 25$
CPU Times of Table 1.1



STEPS TO FOLLOW IN CONSTRUCTING A STEM AND LEAF DISPLAY

Step 1. Divide each observation in the data set into two parts, the stem and the leaf. We will designate the first digit of the CPU time (i.e., the digit to the left of the decimal point) as its stem; we will call the last two digits its leaf. For example, the stem and leaf of the CPU time 2.41 are 2 and 41, respectively:

STEM	LEAF
2	41

Although this assignment is arbitrary, the stem and leaf display will yield more information if you define the stems and leaves so that you obtain a smaller number of stems for a small amount of data and a larger number of stems for large data sets. Using the assignment specified here, we obtain a total of five stems: 0, 1, 2, 3, and 4.

Step 2. List the stems in order in a column, starting with the smallest stem and ending with the largest (see Figure 1.5).

Step 3. Proceed through the data set, placing the leaf for each observation in the appropriate stem row. The completed stem and leaf display for the data of Table 1.1 is shown in Figure 1.5.

Notice that if you rotate the stem and leaf display on its side, you obtain the same type of bar graph as provided by the (relative) frequency distribution. The stem and leaf display of Figure 1.5 partitions the data set into five classes corresponding to the five stems. The number of leaves in each class gives the class frequency.

FIGURE 1.5

Stem and Leaf Display
for the $n = 25$ CPU
Times of Table 1.1

STEMS	LEAVES	FREQUENCY	RELATIVE FREQUENCY
0	15 19 92 82 75 71 47 96 02	9	.36
1	17 23 61 16 94 38 59 40	8	.32
2	41 59 16 01	4	.16
3	76 07 53	3	.12
4	75	1	.04
TOTALS		$n = 25$	1.00

One advantage of a stem and leaf display over a frequency distribution is that the original data are preserved. That is, you can look at the display and resurrect the exact values of the data. A stem and leaf display also arranges the data in an orderly fashion and makes it easy to determine certain numerical characteristics to be discussed in the following sections. The third advantage is that the classes and the numbers falling in them are quickly determined once we have selected the digits that we want to use for the stems and leaves.

A disadvantage of the stem and leaf display is that there is sometimes not much flexibility in choosing the stems. For the data of Table 1.1, two stem and leaf options are possible. We could define the stems and leaves as shown in Figure 1.5. Or, we could let the first two digits represent the stem, in which case the number 2.41 would have the stem 24 and the leaf 1:

STEM	LEAF
24	1

The associated stem and leaf display for the data of Table 1.1 would then contain a total of 48 stems, ranging from 00 to 47, and most of these stem rows would contain no leaves. Clearly, this choice of stems and leaves would not provide as much information about the data as does the display of Figure 1.5. Consequently, we are left with the option of using a stem and leaf display that produces five stems (and thus, five classes for the frequency distribution) or one that produces 48 stems.*

Frequency distributions or relative frequency distributions are most often used in scientific publications to describe quantitative data sets. They are better suited to the description of large data sets and they permit a greater flexibility in the choice of class width.

*By sacrificing some of the simplicity of our procedure, we could define the stems and leaves so that the number of stems falls between 5 and 48. We omit discussion of this topic.

1 SOME BASIC CONCEPTS

Since the z-score is positive, we conclude that the CPU time of 3.76 seconds lies a distance of 1.79 standard deviations above (to the right of) the sample mean of 1.63 seconds. ■

EXERCISES 1.33–1.37

- 1.33 Find the 25th, 50th, and 75th percentiles for the data set in Exercise 1.25.
- 1.34 Refer to Exercise 1.19.
- The 80th percentile for the monthly salaries of the 35 operations analysts is \$2,470. Interpret this value.
 - Find the upper and lower quartiles for the 35 salaries.
 - Find the z-score for the monthly salary of \$1,920. Interpret this value.
- 1.35 The relative frequency distribution of the 144 sample DDT measurements (ppm) (Appendix III) is summarized by the following numerical descriptive measures:
- $$\bar{y} = 24.36 \quad s = 98.38 \quad m = 7.15$$
- Find the 50th percentile for the sample DDT measurements.
 - Use Tchebysheff's theorem to find an upper bound for the 75th percentile of the sample of DDT measurements.
 - Find the z-score for a fish specimen with a DDT measurement of 500 ppm. Interpret this value.
- 1.36 A simulation study was conducted to investigate the rounding accuracies of several new algorithms for functions in the FORTRAN computer program library (*IBM Journal of Research and Development*, Mar. 1986). For each new FORTRAN function, the rounding error for each of 10,000 trials was calculated.
- The 99th percentile of the rounding errors for one particular FORTRAN function was found to be .53. Interpret this value.
 - For the FORTRAN function of part a, the mean rounding error is $\bar{y} = .22$ and the standard deviation (estimated) is $s = .07$. Use this information to calculate the z-score for the 99th percentile value of .53. Interpret the z-score.
- 1.37 Refer to the scram data given in Exercise 1.17.
- Find the upper quartile for the data set. Interpret this value.
 - Find the z-score for a nuclear reactor unit with 9 scrams in 1984. Interpret this value.

SECTION 1.9

METHODS FOR DETECTING OUTLIERS

Sometimes inconsistent observations are included in a data set. For example, when we discuss starting salaries for college graduates with bachelor's degrees, we generally think of traditional college graduates—those near 22 years of age with 4 years of college education. But suppose one of the graduates is a 34-year-old Ph.D. chemical engineer who has returned to the university to obtain a bachelor's degree in metallurgy. Clearly, the starting salary for this graduate could be much larger than the other starting salaries because of the graduate's additional education and experience, and we probably would not want to include it in the data set. Such an errant observation, which lies outside the range of the data values that we want to describe, is called an outlier.

DEFINITION 1.18

An observation y that is unusually large or small relative to the other values in a data set is called an outlier.

The most obvious method for determining whether an observation is an outlier is to calculate its z -score (Section 1.8). For example, if the z -score for a y value is -4.13 , we know that it lies more than 4 standard deviations below the mean of the data set. Both the Empirical Rule and Tchebysheff's theorem tell us that almost all the observations in a data set will have z -scores less than 3 in absolute value. Thus, a z -score as small as -4.13 is highly improbable and points to the possibility of an errant observation.

Another procedure for detecting outliers is to construct a box plot of the data. With this method, we construct intervals similar to the $\bar{y} \pm 2s$ and $\bar{y} \pm 3s$ intervals of the Empirical Rule; however, the intervals are based on a quantity called the interquartile range instead of the standard deviation s .

DEFINITION 1.19

The interquartile range, IQR, is the distance between the upper and lower quartiles:

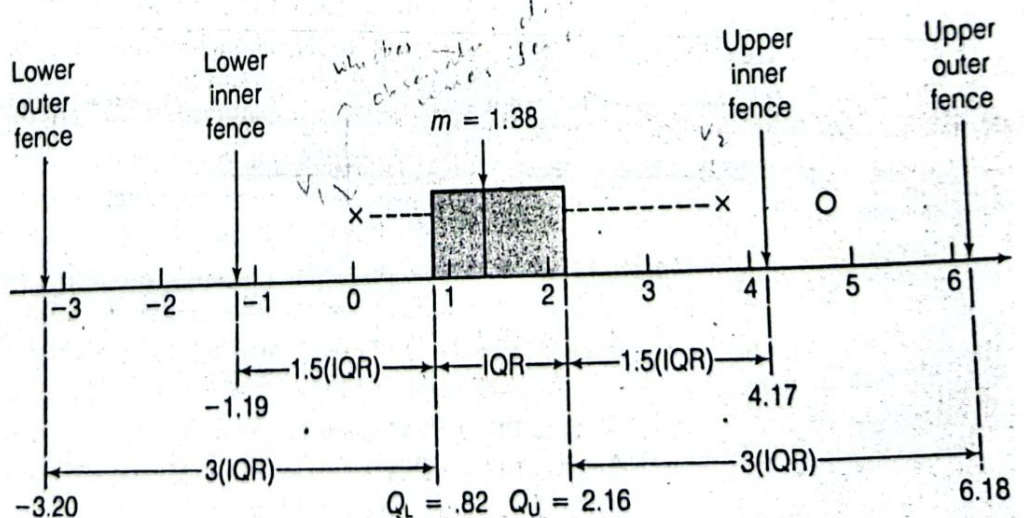
$$IQR = Q_U - Q_L$$

The procedure is especially easy to use for small data sets because the quartiles and interquartile range can be quickly determined. The steps to follow in constructing a box plot are given in the box on page 30.

A box plot for the 25 CPU times in Table 1.1 is shown in Figure 1.7. From the plot you can see that $Q_L = .82$, $m = 1.38$, $Q_U = 2.16$, and

$$IQR = Q_U - Q_L = 2.16 - .82 = 1.34$$

FIGURE 1.7
Box Plot for the $n = 25$
CPU Times of Table 1.1



The inner and outer fences are located a distance of $1.5(IQR) = 1.5(1.34) = 2.01$ and $3(IQR) = 3(1.34) = 4.02$ below Q_L and above Q_U , respectively. Note that the data set contains only one suspect outlier, indicated by the small circle between the upper inner and outer fences. This is the CPU time of 4.75—the largest value in the data set.

The z-score and box plot methods both establish rule-of-thumb limits outside of which a y value is deemed to be an outlier. Usually the two methods produce similar results. However, the presence of one or more outliers in a data set can inflate the value of s used to calculate the z-score. Consequently, it will be less likely that an errant observation would have a z-score larger than 3 in absolute value. In contrast, the values of the quartiles used to calculate the fences for a box plot are not affected by the presence of outliers.

STEPS TO FOLLOW IN CONSTRUCTING A BOX PLOT

- Step 1.** Calculate the median, m , lower and upper quartiles, Q_L and Q_U , and the interquartile range, IQR, for the y values in a data set.
- Step 2.** Construct a box on the y-axis with Q_L and Q_U located at the lower corners (see Figure 1.7). The base width will then be equal to IQR. Draw a vertical line inside the box to locate the median, m .
- Step 3.** Construct two sets of limits on the box plot. Inner fences are located a distance of $1.5(IQR)$ below Q_L and above Q_U ; outer fences are located a distance of $3(IQR)$ below Q_L and above Q_U (see Figure 1.7).
- Step 4.** Observations that fall between the inner and outer fences are called suspect outliers. Locate the suspect outliers on the box plot using small circles. Observations that fall outside the outer fences are called highly suspect outliers. Use solid dots to locate highly suspect outliers.
- Step 5.** To further highlight extreme values, whiskers are added to the box plot. Mark the y value in the region between Q_L and the lower inner fence that is closest to the inner fence with an $\times$, and join the $\times$ to the box with a dashed line—a whisker (see Figure 1.7). Similarly, use an $\times$ and attached whisker to locate the most extreme value between Q_U and the upper inner fence.

EXERCISES 1.38–1.41

- 1.38** Refer to the scam data given in Exercise 1.17.
 - a. Construct a box plot for the data. Do you detect any outliers?
 - b. Use the method of z-scores to detect outliers.
- 1.39** Refer to the voltage reading data for the old location, supplied in Exercise 1.18.
 - a. Construct a box plot for the data. Do you detect any outliers?
 - b. Use the method of z-scores to detect outliers.

- 1.40 Refer to the lead concentration data of Exercise 1.30.
- Construct a box plot for the data. Do you detect any outliers?
 - Use the method of z-scores to detect outliers.
- 1.41 Refer to the 144 DDT measurements in Appendix III.
- Construct a box plot for the data. Do you detect any outliers?
 - Use the method of z-scores to detect outliers.

SECTION 1.10

SAMPLE STATISTICS AND POPULATION PARAMETERS

Numerical descriptive measures computed from sample data are often called statistics. Thus, the sample mean $\bar{y}$ and standard deviation s are examples of statistics. In contrast, numerical descriptive measures of the population are called parameters. Their values are typically unknown and can only be represented by symbols. For example, the population mean and standard deviation are commonly represented by the Greek letters μ and σ , respectively; the population variance is σ^2 . Although we *could* calculate the values of these parameters if we actually had access to the entire population, we generally wish to avoid doing so, for economic or other reasons. Thus, as you will subsequently see, we will *sample* the population and then use the sample statistics to infer, or make decisions about, the values of population parameters.

DEFINITION 1.20

A statistic is a numerical descriptive measure computed from sample data.

DEFINITION 1.21

A parameter is a numerical descriptive measure of a population.

SECTION 1.11

THE ROLE OF STATISTICS IN THE SCIENCES

Experimental scientific research, in engineering, information sciences, and other disciplines, involves the use of experimental data—a sample—to infer the nature of some conceptual population that characterizes a phenomenon of interest to the experimenter. This inferential process is an integral part of the scientific method. Inference based on experimental data is first employed to develop a theory about some phenomenon. Then the theory is tested against additional sample data.

How does the science of statistics contribute to this process? To answer this question, we must note that inferences based on sample data will almost always be subject to error, because a sample will not provide an exact image of the population. The nature of the information provided by a sample depends on the particular sample chosen, and thus will change from sample to sample. For example, suppose you manufacture computers and you want to estimate the proportion of all potential customers who would buy your product. You contract with a sample survey organization to sample 100 potential customers and find